Que 1.	Tower – Simulink	Block -Stateflow

Consider two numbers **a and b**. Design a Stateflow flowchart to perform algebraic operations between a and b. The algebraic operations addition, subtraction, multiplication and division are numbered from 1 to 4 respectively. The user is then asked to select one of the operations he wants to perform. This number is stored in a new variable **choice**.

Depending upon the value of the choice variable (from 1 to 4), the corresponding operation is performed. Find output when a=17, b=6 and choice is subtraction.

Que 2.	Tower – Simulink	Block -Control System

1. Model a DC motor using the equations below.

$$L\frac{di}{dt} = -kw - Ri + V$$
$$J\frac{dw}{dt} = Ki$$

Start with the dc _motor_start model. Note the input voltage V is modeled with a unit step function. After modeling the equations, set the initial conditions for both states to 0.

Parameters-

R-Resistance = 0.3Ω

L-Inductance = $1e^{-6}\ H$

J-Inertia = $6\ Kg - m^2$

K- Torque constant = $0.5\ N - \frac{m}{A}$

k-EMF constant = $0.5\ V - s/rad$

Create a PID controller for speed control using basic block of Simulink (don't used PID block). You can use a unit step as reference speed and the output should track the desired speed.

Que 3.	Tower – Simulink	Block -Stateflow
Design a Stateflow model for a light switch. Consider **ON** and **OFF** states for this logic. When switch is **ON** LightSignal is **ON** (1) and when switch is **OFF** LightSignal is OFF (0).		

Que 4.	Tower – Simulink	Block -Stateflow
Consider two tanks for two tank system. Initially, the levels of both the tanks are 15. The conditions of transition between "**Flow1**" and "**Flow2**" states are specified by the arrows between two modes. The condition *[Level1<=5]* means that the waterflow through pipe will be switched to tank 2 instantaneously whenever the water level of tank 1 falls down to 5. Similarly, the condition *[Level2<=5]* guarantees that the waterflow through pipe will be switched to tank 1 instantaneously whenever the water level of tank 2 falls down to 5. During **Flow1** state, level of tank 1 (Level1) is decremented by 1 level and level of tank 2 (Level2) is incremented by 1 level. During **Flow2** state, level of tank 2 (Level2) is decremented by 1 level and level of tank 1 (Level1) is incremented by 1 level. Design a Stateflow model for this system with given conditions.		

Que 5.	Tower – Simulink	Block -Stateflow
Design a Stateflow model for a washing machine. The input given to this machine is voltage and it has four output indicators i.e., Power, Wash, Rinse, Dry. Following states are required for modelling of this washing machine logic: 1. **OFF:** It is a default state. In this state, the output Power is 0. 2. **ON:** When voltage is given to machine i.e., vtg is 1, this state is active. The output Power is 1 for this state. After 1 second, this machine goes from ON state to Washing state. 3. **Washing:** In this state, output indicator Wash is 1. After 20 seconds, this state exits and enters rinsing state. Output indicator Wash becomes 0. 4. **Rinsing:** In this state, output indicator Rinse is 1. After 10 seconds, this state exits and enters Drying state. Output indicator Rinse becomes 0. 5. **Drying:** In this state, output indicator Dry is 1. After 3 seconds, this state exits, and machine turns OFF. Output indicator Dry becomes 0.		